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Chapter 5: Functions of Random Variables (Summary)

Based on *Probability, Random Variables, and Stochastic Processes* (Papoulis).

Section 5-1: Functions of One Random Variable

This section deals with the problem: Given a random variable (RV) $\mathbf{x}$ with a known distribution, finding the distribution of a new RV $\mathbf{y}$ defined as $\mathbf{y} = g(\mathbf{x})$.

1. The Concept

Given $\mathbf{y} = g(\mathbf{x})$, for any specific value y , the event $\{\mathbf{y} \leq y\}$ is equivalent to the event $\{\mathbf{x} \in R_y\}$, where R_y is the set of x values such that $g(x) \leq y$.

CDF Relation:

$$F_y(y) = P\{\mathbf{y} \leq y\} = P\{\mathbf{x} \in R_y\}$$

2. The Fundamental Theorem (PDF Transformation)

If we want to find the Probability Density Function (PDF) $f_y(y)$ directly from $f_x(x)$, we use the fundamental theorem.

Theorem: To find $f_y(y)$ for a specific value y : 1. Solve the equation $y = g(x)$ for x . 2. Let $x_1, x_2, \dots, x_n, \dots$ be the real roots of this equation ($g(x_i) = y$). 3. Then:

$$f_y(y) = \sum_i \frac{f_x(x_i)}{|g'(x_i)|}$$

Conditions: * If $y = g(x)$ has no real roots, then $f_y(y) = 0$. * $g'(x)$ is the derivative of $g(x)$.

3. Special Cases

A. Linear Transformation If $\mathbf{y} = a\mathbf{x} + b$ (where $a \neq 0$): * There is only one root: $x = (y - b)/a$. * The derivative is $g'(x) = a$.

Formula:

$$f_y(y) = \frac{1}{|a|} f_x\left(\frac{y - b}{a}\right)$$

B. Square Law Transformation If $\mathbf{y} = \mathbf{x}^2$ (where $y > 0$): * There are two roots: $x_1 = \sqrt{y}$ and $x_2 = -\sqrt{y}$. * The derivative is $g'(x) = 2x$. * $|g'(x_1)| = 2\sqrt{y}$ and $|g'(x_2)| = |-2\sqrt{y}| = 2\sqrt{y}$.

Formula:

$$f_y(y) = \frac{1}{2\sqrt{y}} [f_x(\sqrt{y}) + f_x(-\sqrt{y})] \quad \text{for } y > 0$$

4. Expectation of a Function

Instead of finding $f_y(y)$ first to calculate the mean, we can compute it directly using $f_x(x)$.

Theorem:

$$E[g(\mathbf{x})] = \int_{-\infty}^{\infty} g(x) f_x(x) dx$$

Section 5-2: Functions of Two Random Variables

This section extends the concept to two RVs: Given $\mathbf{x}$ and $\mathbf{y}$ with joint density $f_{xy}(x, y)$, find the distribution of $\mathbf{z} = g(\mathbf{x}, \mathbf{y})$.

1. One Function of Two Variables

$$\mathbf{z} = g(\mathbf{x}, \mathbf{y})$$

General Method (CDF):

$$F_z(z) = P\{\mathbf{z} \leq z\} = \iint_{D_z} f_{xy}(x, y) dx dy$$

* Where D_z is the region in the xy -plane such that $g(x, y) \leq z$.

Important Specific Functions: A. Sum of Two RVs (Convolution) If $\mathbf{z} = \mathbf{x} + \mathbf{y}$: The density $f_z(z)$ is found by integrating along the line $x + y = z$.

$$f_z(z) = \int_{-\infty}^{\infty} f_{xy}(x, z - x) dx$$

- **Corollary:** If $\mathbf{x}$ and $\mathbf{y}$ are **independent**, $f_{xy}(x, y) = f_x(x)f_y(y)$. The formula becomes the **Convolution Integral**:

$$f_z(z) = f_x(z) * f_y(z) = \int_{-\infty}^{\infty} f_x(x)f_y(z - x) dx$$

B. Ratio of Two RVs If $\mathbf{z} = \mathbf{x}/\mathbf{y}$:

$$f_z(z) = \int_{-\infty}^{\infty} |y| f_{xy}(yz, y) dy$$

2. Two Functions of Two Variables (Transformation)

Given two new variables defined by:

$$\mathbf{z} = g(\mathbf{x}, \mathbf{y})$$

$$\mathbf{w} = h(\mathbf{x}, \mathbf{y})$$

To find the joint density $f_{zw}(z, w)$:

The Fundamental Theorem (2D):

1. Solve the system for x and y in terms of z and w .

2. Let $(x_1, y_1), (x_2, y_2), \dots$ be the real roots.
3. Calculate the **Jacobian** of the transformation $J(x, y)$:

$$J(x, y) = \det \begin{vmatrix} \frac{\partial z}{\partial x} & \frac{\partial z}{\partial y} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} \end{vmatrix} = \frac{\partial z}{\partial x} \frac{\partial w}{\partial y} - \frac{\partial z}{\partial y} \frac{\partial w}{\partial x}$$

Formula:

$$f_{zw}(z, w) = \sum_i \frac{f_{xy}(x_i, y_i)}{|J(x_i, y_i)|}$$

3. The Auxiliary Variable Method

If you are given only one function $\mathbf{z} = g(\mathbf{x}, \mathbf{y})$ but want to find $f_z(z)$ using the Jacobian method: 1. Introduce an arbitrary auxiliary variable, usually $\mathbf{w} = \mathbf{x}$ (or $\mathbf{w} = \mathbf{y}$). 2. Find the joint density $f_{zw}(z, w)$ using the Fundamental Theorem above. 3. Integrate out $\mathbf{w}$ to get the marginal density of $\mathbf{z}$:

$$f_z(z) = \int_{-\infty}^{\infty} f_{zw}(z, w) dw$$

5. Classic Textbook Examples (Must-Know)

These are the standard examples from Papoulis (Chapter 5) that illustrate the core theorems.

Example 5-2: The Square Law Device (Transformation of One RV)

Problem: Consider a device with input $\mathbf{x}$ and output $\mathbf{y}$ defined by the square-law characteristic:

$$\mathbf{y} = a\mathbf{x}^2 \quad (a > 0)$$

Find the PDF $f_y(y)$ given $f_x(x)$.

Solution: 1. **Analyze the Roots:** * For $y < 0$: The equation $y = ax^2$ has no real solutions. Thus, $f_y(y) = 0$. * For $y > 0$: The equation has two solutions (roots):

$$x_1 = \sqrt{\frac{y}{a}}, \quad x_2 = -\sqrt{\frac{y}{a}}$$

2. **Compute Derivatives:** * $g'(x) = 2ax$. * At the roots: $|g'(x_1)| = |2a\sqrt{y/a}| = 2\sqrt{ay}$ and $|g'(x_2)| = |-2a\sqrt{y/a}| = 2\sqrt{ay}$. 3. **Apply Fundamental Theorem:**

$$f_y(y) = \frac{f_x(x_1)}{|g'(x_1)|} + \frac{f_x(x_2)}{|g'(x_2)|}$$

$$f_y(y) = \frac{1}{2\sqrt{ay}} \left[f_x\left(\sqrt{\frac{y}{a}}\right) + f_x\left(-\sqrt{\frac{y}{a}}\right) \right] u(y)$$

Special Case (Chi-Square Degree 1): If $\mathbf{x} \sim N(0, 1)$ (Standard Normal) and $a = 1$: * $f_x(x)$ is even, so $f_x(\sqrt{y}) = f_x(-\sqrt{y})$. * The formula simplifies to:

$$f_y(y) = \frac{1}{\sqrt{y}} f_x(\sqrt{y}) = \frac{1}{\sqrt{2\pi y}} e^{-y/2} u(y)$$

* This is the **Chi-Square distribution with 1 degree of freedom**.

Example 5-12: Sum of Independent Uniform RVs (Transformation of Two RVs)

Problem: Let $\mathbf{z} = \mathbf{x} + \mathbf{y}$. Find the PDF $f_z(z)$ if $\mathbf{x}$ and $\mathbf{y}$ are independent and uniformly distributed in $(0, 1)$.

Solution: 1. **Identify the Method:** Since $\mathbf{x}$ and $\mathbf{y}$ are independent, the PDF of their sum is the **convolution** of their individual PDFs:

$$f_z(z) = f_x(z) * f_y(z) = \int_{-\infty}^{\infty} f_x(z-y)f_y(y) dy$$

2. **Setup the Integral:** * $f_x(x) = 1$ for $0 < x < 1$, else 0. * $f_y(y) = 1$ for $0 < y < 1$, else 0. * The integrand $f_x(z-y)f_y(y)$ is non-zero only when both $0 < y < 1$ AND $0 < z-y < 1$ (which means $z-1 < y < z$). 3. **Evaluate Ranges:** The intersection of intervals $(0, 1)$ and $(z-1, z)$ depends on z : * **Case 1** ($0 \leq z < 1$): Overlap is $(0, z)$.

$$f_z(z) = \int_0^z 1 \cdot 1 dy = z$$

* **Case 2** ($1 \leq z < 2$): Overlap is $(z-1, 1)$.

$$f_z(z) = \int_{z-1}^1 1 \cdot 1 dy = 1 - (z-1) = 2 - z$$

* **Case 3 (Otherwise):** No overlap, integral is 0. 4. **Result (Triangular Distribution):**

$$f_z(z) = \begin{cases} z & 0 \leq z < 1 \\ 2 - z & 1 \leq z < 2 \\ 0 & \text{otherwise} \end{cases}$$

* (*Tip: The convolution of two rectangular pulses is always a triangular pulse.*)

Summary of Key “Tricks” for Solving Problems

1. **Symmetry Trick:** If $f_x(x)$ is even and $y = x^2$, you can just compute for positive x and multiply by 2 (as seen in Example 5-2).
2. **Auxiliary Variable Trick:** When finding the distribution of $\mathbf{z} = g(\mathbf{x}, \mathbf{y})$, if the Jacobian method seems too complex, consider finding the CDF $F_z(z)$ by integrating over the region D_z in the xy -plane directly. This is often geometric (calculating areas).
3. **Independence Shortcut:** Always check if $\mathbf{x}$ and $\mathbf{y}$ are independent. If yes, $f_{xy} = f_x f_y$, which simplifies double integrals significantly.